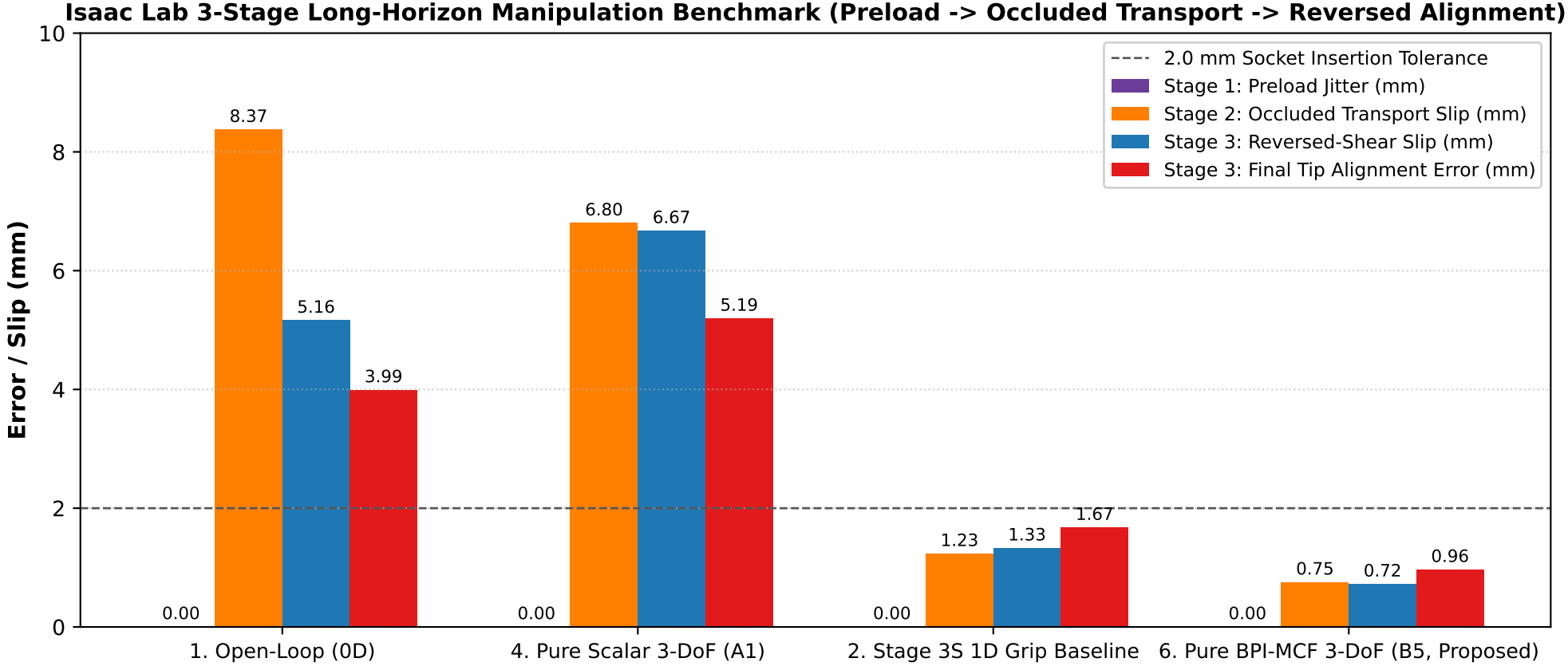
Isaac Lab 3-Stage Long-Horizon Manipulation Benchmark (Preload -> Occluded Transport -> Reversed Alignment)
Error / Slip (mm)
10
8
6
4
2
0
2.0 mm Socket Insertion Tolerance
Stage 1: Preload Jitter (mm)
Stage 2: Occluded Transport Slip (mm)
Stage 3: Reversed-Shear Slip (mm)
Stage 3: Final Tip Alignment Error (mm)
0.00
8.37
5.16
3.99
0.00
6.80
6.67
5.19
0.00
1.23
1.33
1.67
0.00
0.75
0.72
0.96
1. Open-Loop (0D)
4. Pure Scalar 3-DoF (A1)
2. Stage 3S 1D Grip Baseline
6. Pure BPI-MCF 3-DoF (B5, Proposed)